3 - b

Question 1.1

As the number of training points increases, the posterior over the parameters (w_0, w_1) becomes increasingly concenerated and elongated along a specific direction. With no data, the posterior coincides with the isotropic prior. With a few data points, the posterior becomes elongated, which shows that many parameter values can still explain the data. As more data are added, the posterior becomes smaller and more concentrated around the true parameters.

Question 1.5

The predictive variance is minimal inside the training region and increases when moving away from it. This happens because predictions outside the training distribution rely on extrapolation, where the model parameters are less constrained by the data, which leads to higher epistemic uncertainty.

Analytically, for Bayesian linear regression with $\phi(x) = (1, x)$, $\alpha \to 0$ and $\beta = 1$, the predictive variance is $\sigma^2_{pred}(x) = 1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}$

where $\bar{x}$ is the mean of the training inputs and $S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2$

This expression shows that the predictive variance grows quadratically with the distance between x and $\bar{x}$, which explains why uncertainty increases far from the training distribution.

Bonus

Bayes Linear Regression still learns one global straight line that connects the two separated clusters. The predictive uncertainty does not depend on whether there are data points locally, but mainly on how far the inputs is from the training mean. As a result, the model may look confident in the middle gap, even though no data are present there, while uncertainty increases more clearly outside the training regions.

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Question 2.2

The predictive variance is low inside the training region where data are available and increases sharply outside this region. With polynomial basis functions, the variance grows very fast near the boundaries, which shows that the model becomes highly uncertain when extrapolating beyond the training domain.

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